

FINSECURE IT SYSTEM AUDIT REPORT

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1. Audit Objectives

Based on Section 1: Objective of the Lab.

The objective of this audit was to transition from manual to automated compliance checks for FinSecure. The focus is on three key security pillars to protect sensitive financial and personal data:

- **Data Confidentiality:** Enforcing strict file system permissions.
- **Access Control:** Monitoring active user sessions for policy violations.
- **System Availability:** Monitoring disk utilization to prevent service outages.

2. Automated Compliance Functions

Based on Section 2: Explanation of Script Functions.

Function	Security Logic	Technical Implementation
<code>check_file_permissions()</code>	Enforces Least Privilege on sensitive data directories.	Uses <code>os.stat()</code> to ensure <code>/secure_data</code> restricts Group and Others access.
<code>check_active_sessions()</code>	Detects anomalous or excessive concurrent user access.	Utilizes <code>psutil.users()</code> to compare session counts against the threshold.
<code>check_disk_space()</code>	Ensures storage is sufficient for continuous operations.	Uses <code>psutil.disk_usage('/')</code> to alert if utilization exceeds the defined limit.

3. Audit Results & Validation

Based on Section 4: Testing and Results.

Validation was conducted by adjusting script thresholds to simulate non-compliant environments. Key findings include:

- **Permission Integrity:** The script correctly identifies `stat.S_IRWXO` violations, triggering an incident response when permissions are too permissive.

- **Session Monitoring:** Concurrent access over the policy limit (e.g., more than 2 sessions) is flagged as "NOT Compliant".
- **Storage Safety:** Low thresholds correctly identify disk exhaustion risks, ensuring high availability of financial services.

4. Conclusion & Recommendations

Based on Section 5: Conclusion.

By implementing Python-based automation, FinSecure significantly reduces human error and audit time. It is recommended to integrate this script into a continuous monitoring dashboard for real-time alerting on potential security misconfigurations.